EGA V, English translation by P. Blass, 1987 (cote 24)

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HYPERPLANE SECTIONS AND CONIC PROJECTIONS

Section 1: Preliminaries and Notation.

Let S be a prescheme, let $\underline{E}$ be a locally free module of finite type over S let $\underline{E}^\vee$ be its dual. We denote by $\mathrm{P} = \underline{P}(\underline{E})$ the projective fibration defined by $\underline{E}$ and by $\mathrm{P}^\vee$ the projective fibration defined by $\mathrm{E}^\vee$. We shall call $\mathrm{P}^\vee$ the scheme of hyperplanes of P. This terminology can be justified as follows. Let ξ be a section of $\mathrm{P}^\vee$ over S which is therefore determined by an invertible quotient module L of $\underline{E}^\vee$. From it we obtain an invertible quotient L_P of $(\underline{E}^\vee)_{\underline{P}} = (\underline{E}_P)^\vee$, on the other hand, we have the invertible quotient module $\mathrm{O}_P(1)$ of E_P. Passing to duals we may take $\underline{L}_P{-}1$ resp. $\underline{O}_P(-1)$ to be invertible submodules (locally direct summands or factors) of $\underline{E}_P$ (resp. of $(\underline{E}_P)^\vee$) and the pairing $EP \otimes E_P^\vee \to \underline{O}_P$ defines therefore a natural pairing (*) $\underline{O}_P(-1) \otimes \underline{L}_P{-}1 \to \underline{O}_P$ or also the transpose homomorphism (**) $\underline{O}_P \to \underline{O}_P(1) \otimes \underline{L}_P = \underline{O}_P(1)$ i.e. a section of $\underline{L}_P(1)$ canonically defined by ξ. The "divisor" of that section i.e. the closed subscheme H_ξ of $\underline{P}$ defined by the image ideal of (*) is called the hyperplane in $\underline{P}$ defined by the element $\xi \in \mathrm{P}^\vee(\mathrm{S})$. We could describe it by noting that locally over $\underline{S}$, ξ is given by a section ϕ of E such that ϕ (s) $\neq 0$ for all s (ϕ is determined by ξ up to multiplication by an invertible section of $\underline{O}_S$); since $\mathrm{E} = \mathrm{p}^*(\mathrm{O}_P(1))$ (p: P $\to$ S being the projection), ϕ can be considered as a section of $\mathrm{O}_P(1)$ the divisor of which is nothing else but H_ξ.

Of course, if we consider L-1 as an invertible submodule of E locally a direct factor in E then the correspondence between ξ

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not be immersed it is necessary and sufficient that Z_i should be such [PB incomprehensible].* If that is so then the radicial multiplicity of F at Z_i at k is equal to that of G_η at Z_i relative to $k(\eta)$.

The first assertion is contained in the last assertion of 2.7. For the second, since $Y_\eta \to X$ is flat therefore the functor $F \to G_\eta$ is exact, we are reduced by restriction to a neighborhood of the general point of Z_i and by a devissage (unscrewing) to the case where $F = O_{Z_i}$ and we may assume $Z_i = X$. Also we could start by assuming that X is separable over k is reduced to the case of k algebraically closed [PB: incomprehensible].* Then the assertion is contained in 2.15. (iii). Then we conclude as usual:

Corollary 2.17

Let Z be the set of points of X where F is not separable over k (reference). Then G_η is separable over $k(\eta)$ if and only if Z is finite. In particular if F is separable over k then G_η is separable over $k(\eta)$.

Remark 2.18

The key result 2.12 (and its consequences 2.13 and 2.17) become false if we abandon the assumption that f is an immersion as we see for example by taking for X a purely inseparable covering of $\underline{P}$. However if k is of characteristic zero 2.12 and 2.17 are valid without assuming that f is an immersion.

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have dim f(X_i) > [illegible] [is it two ask Grothendieck]* then Z is of codimension two in $\underline{P}$v. This follows from 2.1 and 2.2. (which implies that every irreducible component of $\underline{Y}$ dominates $\underline{P}$v) and from the dimension theory for the morphism $\underline{Y} \to \underline{P}$v. Starting from this result we may give as a corollary the case where we start from a closed subset Z of X and where we consider the dimension of the inverse images Z_ξ in the Y_ξ ($\xi \in \underline{P}$v) and we may even take for Z the set of $\xi \in \underline{P}$v such that there exists an irreducible component of $T_{k(\xi)}$ whose trace on Y_ξ has the greatest dimension (NB we always assume that for every irreducible Z_i of Z we have dim f(Z_i) > 0).

Finally the most precise statement in this direction and one that results easily from the first announcement (for X irreducible) and from 2.7 is the following modified statement: F being coherent over X, suppose that for every associated prime cycle T for F we have dim f(T) > 0 then the set of $\xi \in \underline{P}$v such that ϕ_ξ is not $F_{k(\xi)}$ - regular is (constructible and) of codimension ≥ 2. (The notation for ϕ_ξ is that from N05). We can give this as the principal assertion, and announce the previous assertions as corollaries, the proof being or proceeding via one of the corollaries.

Please note that with the preceding notations if $\xi \in \overset{\text{v}}{\underline{P}}$ - Z, then for every y $\in \underline{Y}_\xi$ we have coprof$_y$ $G_{k(\xi)}$ = coprof$_y$ G_ξ and consequently if coprof F $\leq$ n in particular if F is Cohen-Macauley then for $\xi \in \underline{P}$v - Z, G_ξ is Cohen-Macauley. Finally if F is (S_k) [illegible ask Grothendieck]*

S_{k-1} for $\xi \in \underline{P}$v - Z (reference O_{IV}).

{marg.: * Ask AG.}